2.9 Appendices

2.9.1 Appendix 1

Selection of the predictor variables for the shallow landslide susceptibility assessment

Checking for multicollinearity between the independent variables

Multicollinearity occurs when two or more predictor variables are strongly correlated with each other. In other words, multicollinearity occurs when one or more predictor variables can be accurately predicted by a linear combination of one or more of the other variables. Multicollinearity between predictor variables will not influence the accuracy or quality of the final landslide model. However, it can bias our interpretation of the variable importance. Hence, it is recommended to remove any strong multicollinearity between the predictor variables prior to constructing and analyzing the landslide model. Here we calculated two techniques to assess multicollinearity: the Variance Inflation Factor (VIF) and the Condition Indices (Cis).

The VIF for a predictor variable i is calculated as $\frac{1}{1-R^2}$

Whereby the R2 is derived from a regression model that predicts variable i with the other variables as predictor. We consider a variable problematic (in terms of multicollinearity) when its VIF>4 (Belsley et al., 2005). For the variables used in our landslide susceptibility model, we observed no problematic VIF (Table 10).

Table 10: **Variance inflation factor for the continuous variables used in the landslide susceptibility model.**

Variable	VIF
Slope	1.04
Elevation	1.73
Profile curvature	1.17
Planar curvature	1.20
North exposure	1.00
East exposure	1.00
PGA	2.96
Distance to rivers	1.16
Distance to faults	3.46
Road (50m buffer)	1.00
Soil with >35% clay	1.06

Note that the morphological parameters were derived here from the 10 m resolution DEM.